



Nb₂CT_x MXene — Single-/Few-Layer Powder

Research-grade two-dimensional transition-metal carbide powder

Nb₂CT_x MXene is supplied as a delaminated single-/few-layer powder derived from Nb₂AlC by HF etching followed by intercalation and delamination. The material has a layer count of ≤5, nominal lateral size of 200–1000 nm, typical conductivity of 10–100 S/cm, and mixed –O, –OH and –F surface terminations.

KEY FEATURES

- Single-/few-layer powder, ≤5 layers
- Nominal lateral size: 200–1000 nm
- Typical conductivity: 10–100 S/cm
- Mixed –O/–OH/–F surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

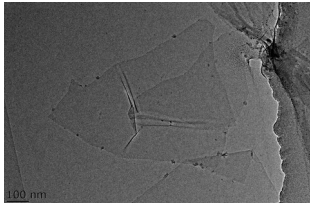
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Nb ₂ CT _x MXene
Precursor	Nb ₂ AlC
Product form	Delaminated single-/few-layer powder
Preparation	HF etching followed by intercalation / delamination
Layer structure / count	≤5 layers
Nominal lateral size	200–1000 nm
Typical conductivity	10–100 S/cm
Surface terminations	–O, –OH, –F
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

	Single-/few-layer Nb ₂ CT _x — Representative microscopy. Representative characterization; not lot-specific. Technique not specified.
	Single-/few-layer Nb ₂ CT _x — XRD. Representative characterization; not lot-specific. Non-data annotation removed for presentation; diffraction trace and axes otherwise unchanged.

Product positioning: Supplied and technically supported by WiniGen Materials for research and development use.